

HumanEval typicality metrics — qwen3.5-9b & gemma-4-31b-it (2026-06-16)

Settings: s2 = RankAlign; s4 = New+fsx+self-tc (FLORA-PMI); base = untrained qwen. Eval: basetyp = self-typ + base-typ; self = self-typ no-base; basetypneg = neg-typ + base-typ; neg = neg-typ no-base. Gen variant = tc (typicality-corrected). Metrics averaged over the 82 humaneval test tasks (gemma multi-s4-basetyp: 70, see /datastor1-full note).

Table 1: HumanEval typicality metrics (gen variant = tc): gen ROC, val ROC, val acc, Pearson(gen,val).

Model	Data	Setting	Eval	gen ROC	val ROC	val acc	Pearson
gemma-4-31b-it (trained)	multi	s2	basetyp	0.9068	0.8995	0.7646	0.6486
gemma-4-31b-it (trained)	multi	s2	basetypneg	0.9424	0.8995	0.7646	0.7070
gemma-4-31b-it (trained)	multi	s2	neg	0.8242	0.8995	0.7646	0.5322
gemma-4-31b-it (trained)	multi	s2	self	0.8563	0.8995	0.7646	0.5834
gemma-4-31b-it (trained)	multi	s4	basetyp	0.9013	0.9437	0.8264	0.6902
gemma-4-31b-it (trained)	multi	s4	self	0.8404	0.9437	0.8264	0.5763
gemma-4-31b-it (trained)	upper	s2	basetyp	0.8689	0.9142	0.7683	0.4101
gemma-4-31b-it (trained)	upper	s2	basetypneg	0.8822	0.9142	0.7683	0.5554
gemma-4-31b-it (trained)	upper	s2	neg	0.8626	0.9142	0.7683	0.5993
gemma-4-31b-it (trained)	upper	s2	self	0.8516	0.9142	0.7683	0.4119
gemma-4-31b-it (trained)	upper	s4	basetyp	0.9070	0.9306	0.8415	0.6328
gemma-4-31b-it (trained)	upper	s4	self	0.8433	0.9306	0.8415	0.5254
qwen3.5-9b (trained)	multi	s2	basetyp	0.9307	0.8479	0.7786	0.6473
qwen3.5-9b (trained)	multi	s2	basetypneg	0.9651	0.8480	0.7786	0.7006
qwen3.5-9b (trained)	multi	s2	neg	0.8032	0.8479	0.7786	0.7523
qwen3.5-9b (trained)	multi	s2	self	0.9021	0.8476	0.7786	0.6132
qwen3.5-9b (trained)	multi	s4	basetyp	0.8985	0.8911	0.8050	0.6596
qwen3.5-9b (trained)	multi	s4	self	0.8294	0.8912	0.8046	0.5651
qwen3.5-9b (trained)	upper	s2	basetyp	0.8725	0.8328	0.7431	0.5532
qwen3.5-9b (trained)	upper	s2	basetypneg	0.9022	0.8323	0.7431	0.6948
qwen3.5-9b (trained)	upper	s2	neg	0.7934	0.8326	0.7431	0.6891
qwen3.5-9b (trained)	upper	s2	self	0.8217	0.8326	0.7431	0.4596
qwen3.5-9b (trained)	upper	s4	basetyp	0.8323	0.8633	0.7645	0.5886
qwen3.5-9b (trained)	upper	s4	self	0.7858	0.8630	0.7653	0.5149
qwen3.5-9b (BASE)	multi	base	neg	0.5987	0.8345	0.6703	0.2935
qwen3.5-9b (BASE)	multi	base	self	0.6902	0.8344	0.6684	0.2715
qwen3.5-9b (BASE)	upper	base	neg	0.6573	0.8170	0.6528	0.3042
qwen3.5-9b (BASE)	upper	base	self	0.6907	0.8172	0.6528	0.2648